Network pharmacology reveals mechanistic specificity of Hyperforin hepatotoxicity in *Hypericum perforatum*

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Abstract

Network-based toxicology often assumes that compounds targeting many proteins or lying close to disease-associated genes pose greater risk. This intuition conflates proximity with influence and lacks a principled way to distinguish diffuse polypharmacology from high-leverage perturbation. Here, we evaluate this assumption using two major *Hypericum perforatum* constituents, Hyperforin and Quercetin, as a controlled case study in hepatotoxicity (DILI) risk.

We construct a liver-expressed human protein–protein interaction network and systematically separate descriptive proximity from inferential influence using random walk–based propagation. Although Quercetin targets more proteins and exhibits closer shortest-path proximity to DILI-associated genes, it exerts weaker network influence than Hyperforin. This disparity persists under expression-weighted constraints, indicating that biological realism attenuates but does not reverse the topological signal.

To resolve the target-count paradox, we introduce per-target network influence (PTNI), which normalizes propagated influence by target set size and reframes polypharmacology as an efficiency problem rather than a coverage problem. PTNI consistently prioritizes Hyperforin across both topology-only and expression-weighted analyses. Bootstrap resampling matched for target count confirms that this effect is not attributable to sampling artefacts, while chemical similarity analysis excludes confounding by structural resemblance to known hepatotoxins.

Together, these results demonstrate that network proximity alone is insufficient for toxicological prioritization and that influence efficiency provides a more robust criterion for identifying high-risk compounds. This framework generalizes beyond *H. perforatum* and offers a principled basis for integrating

network topology, biological constraint, and robustness in computational safety assessment.

Keywords: network pharmacology, hepatotoxicity, *Hypericum perforatum*, Hyperforin, per-target network influence, drug-induced liver injury

1 Introduction

Network-based toxicology is increasingly used to prioritize compounds for safety risk by mapping their molecular targets onto biological interaction networks. A common heuristic underlying these approaches is that compounds interacting with many proteins, or lying in close network proximity to disease-associated genes, are more likely to pose adverse risk. Though intuitive, this assumption conflates topological proximity with functional influence and provides no principled way to distinguish diffuse polypharmacology from high-leverage perturbation. Whether proximity and target count meaningfully predict toxicological risk remains an open and testable question.

Hypericum perforatum (St. John's Wort) provides a controlled system in which to examine this assumption. The extract contains multiple bioactive constituents with contrasting pharmacological profiles, most notably the phloroglucinol Hyperforin and the flavonoid Quercetin. Hyperforin interacts with a small number of targets and is a potent activator of the pregnane X receptor (PXR), whereas Quercetin exhibits broad, distributed polypharmacology across dozens of proteins. Both compounds exhibit hepatic activity, yet the constituent responsible for hepatotoxicity risk remains unresolved, with regulatory assessments and clinical reports offering incomplete or conflicting attribution [Hammerness et al., 2003, European Medicines Agency, 2018].

Existing network studies would predict greater risk for Quercetin based on its larger target set and closer proximity to liver injury–associated genes. However, such predictions implicitly assume that all targets contribute equally and that network distance is a sufficient surrogate for downstream perturbation. These assumptions ignore the topology of signal propagation within biological networks, where influence depends not only on distance, but also on how signals traverse highly connected and regulatory nodes.

Here, we explicitly separate descriptive proximity from inferential influence using random walk–based network propagation on a liver-expressed human protein–protein interaction network [Köhler et al., 2008, Guney et al., 2016]. Random walk with restart (RWR) captures the extent to which perturbations originating at compound targets diffuse through the network to a defined disease module, providing an influence-based metric that is not reducible to shortest-path distance alone. To incorporate biological realism, we further constrain propagation by liver-specific gene expression.

To resolve the target-count paradox—where compounds with more targets may exert weaker functional effects—we introduce per-target network influence (PTNI), a normalization that quantifies influence efficiency rather than raw coverage. PTNI reframes polypharmacology as an efficiency problem, distinguishing compounds that weakly perturb many nodes from those that exert strong leverage through a limited set of targets.

We apply this framework to compare Hyperforin (10 targets) and Quercetin (62 targets) across multiple analytical tiers, including proximity, influence, expression-weighted influence, PTNI, bootstrap sensitivity analysis, and chemical similarity controls. This design allows us to test whether target count and proximity are sufficient predictors of hepatotoxic risk, or whether influence efficiency provides a more robust criterion for toxicological prioritization.

2 Results

2.1 Quercetin targets are closer to DILI genes but exert weaker influence

We first established network context by quantifying target count and shortest-path proximity to 82 DILI-associated genes (Figure 1). Quercetin engages 62 targets in the liver-expressed largest connected component; Hyperforin engages 10. Shortest-path proximity reveals that Quercetin targets are significantly closer to DILI genes than random expectation (Z = -5.4; $p < 10^{-6}$; $n = 1{,}000$ degree-matched permutations), while Hyperforin targets exhibit weaker proximity (Z = -2.6; $p = 0.005$). By conventional interpretation, Quercetin should pose greater hepatotoxicity risk.

Random walk with restart (RWR) reverses this ranking (Figure 2). Hyperforin achieves influence Z = +10.3 ($p < 10^{-16}$); Quercetin achieves Z = +4.4 ($p < 10^{-6}$). Both are significant; the magnitudes differ by a factor of 2.3 in Z-score space. Despite being topologically more distant from DILI genes, Hyperforin exerts substantially greater network influence.

This dissociation between proximity and influence—the proximity paradox—demonstrates that shortest-path distance is not a reliable surrogate for functional impact.

2.2 Expression weighting attenuates but does not reverse the advantage

To assess whether the RWR signal persists under tissue-specific biological constraint, we applied expression-weighted influence propagation (EWI), weighting network transitions by liver gene expression (Figure 3).

The Z-score differential narrows under expression weighting: ΔZ = +5.8 (RWR) $\rightarrow$ +3.5 (EWI).

Decomposition reveals the mechanism: Hyperforin loses 1.2 Z-score units due to attenuation of high-expression targets (CYP2C9, CYP3A4), while Quercetin gains 1.1 units, driven primarily by complement factor B (CFB) at 1,115 TPM.

The residual gap remains significant: Hyperforin Z = +9.1 ($p < 10^{-16}$); Quercetin Z = +5.6 ($p < 10^{-8}$). Expression weighting did not manufacture the signal; it refined it. Attenuation is expected when propagation is constrained to biologically active nodes; it does not constitute invalidation.

2.3 Per-target efficiency reveals a 22-fold disparity

To resolve the target-count paradox—where compounds with more targets may exert weaker functional effects—we computed per-target normalized influence (PTNI = total influence / target count), reframing polypharmacology as an efficiency problem rather than a coverage problem (Figure 4).

Compound	**Targets**	**PTNI (RWR)**	**PTNI (EWI)**	**Ratio**
Hyperforin	10	0.0114	0.0133	—
Quercetin	62	0.0005	0.0008	—
Fold difference	—	22×	17×	—

Each Hyperforin target contributes 17–22× more DILI-directed influence than each Quercetin target. This disparity is consistent across both topology-only (RWR) and expression-weighted (EWI) analyses, indicating that the efficiency advantage is robust to biological constraint.

PTNI is a derived descriptive metric; it was not subjected to independent permutation testing. Its utility lies in normalizing for target count, not in providing additional inference.

2.4 Bootstrap resampling excludes target-count confounding

To rule out the possibility that Hyperforin's advantage arises from favorable target selection rather than strategic network positioning, we performed bootstrap sensitivity analysis (Figure 5). 100 random 10-target subsets were sampled from Quercetin's 62-target pool and scored by RWR.

Hyperforin's observed influence (0.114) exceeds the entire bootstrap distribution. The 95th percentile of the Quercetin bootstrap (0.054) falls below Hyperforin's score. The fold difference between Hyperforin and the bootstrap mean is 3.7×.

This confirms that Hyperforin's advantage is not an artifact of target count. Even when sampling equalized subsets from Quercetin's pool, no configuration matches Hyperforin's influence. Bootstrap

analysis is a robustness control; it strengthens confidence in the ranking without providing independent evidence.

2.5 Threshold robustness

The ranking is stable across network confidence thresholds (Table 6). At STRING confidence $\geq$700, Hyperforin achieves RWR Z = +10.9 and EWI Z = +10.5; at $\geq$900, Z = +10.3 and +9.1 respectively. Quercetin ranks second in all configurations. This confirms that the core finding is not sensitive to network construction parameters.

2.6 Chemical similarity excludes structural confounding

To exclude the possibility that Hyperforin's network signal reflects structural similarity to known hepatotoxins, we performed chemical similarity analysis against the DILIrank reference set (Figure 6). Morgan fingerprints (ECFP4, radius 2, 2048 bits) were computed for Hyperforin and Quercetin, and maximum Tanimoto similarity to 542 DILI-positive and 365 DILI-negative drugs was determined.

Compound	**Max (DILI+)**	**Max (DILI−)**	**Analog?**
Hyperforin	0.15	0.20	No
Quercetin	0.21	0.22	No

Neither compound exceeds the 0.4 threshold for structural analog detection [Maggiora et al., 2014]. Notably, Quercetin exhibits higher structural similarity to DILI reference drugs yet lower network influence, reinforcing that the observed asymmetry is not driven by chemical features.

Chemical similarity analysis is an orthogonal exclusion criterion, not a risk predictor. It rules out structural confounding; it does not establish causation.

3 Discussion

3.1 Chemical similarity control

Chemical similarity analysis excludes the possibility that either compound is a structural analog of known hepatotoxins. Maximum Tanimoto similarity to DILIrank reference drugs falls below the 0.4 threshold for both compounds. This rules out structural resemblance as an alternative explanation for the observed network signal. It does not imply that either compound is structurally inert.

3.2 Bootstrap robustness

Bootstrap resampling eliminates target-count as a sufficient explanation for Hyperforin's ranking. No random 10-target subset from Quercetin's pool matches Hyperforin's observed influence. This strengthens confidence in the ranking without adding independent evidence. The bootstrap is a robustness control, not a confirmatory test.

3.3 Per-target network influence

PTNI reframes polypharmacology as an efficiency problem rather than a coverage problem. Compounds with fewer targets may exert higher per-target influence if those targets occupy topologically central positions. PTNI is a conceptual normalization; it was not subjected to permutation testing and should not be interpreted as a hypothesis test. Its utility lies in resolving the target-count paradox, not in providing additional inference.

3.4 Expression-weighted influence

Expression weighting modulates but does not rescue proximity-based intuition. Quercetin gains influence under liver-specific expression constraints; Hyperforin's advantage narrows but persists. Attenuation is expected when expression reweights signal propagation. Attenuation does not constitute invalidation.

3.5 The proximity–influence paradox

Proximity counts shortest paths; influence weights them by topology. These are distinct properties. The rank reversal observed here—closer proximity, weaker influence—challenges a common heuristic in network medicine: that targets closer to disease genes are more relevant. This assumption is convenient but incomplete. Influence-based metrics, while computationally more demanding, are not reducible to distance.

3.6 Generalizability

The framework generalizes beyond this case study. Any polypharmacological comparison faces the same confounds: target count, proximity, and expression context. PTNI, expression weighting, and robustness controls address these systematically. The framework is a template, not a validation of Hyperforin hepatotoxicity. What generalizes: the analytical structure. What does not generalize: the specific ranking,

which is contingent on data sources and network construction.

3.7 Limitations

Drug–target databases may be incomplete; novel targets could alter the ranking. STRING aggregates interactions across cell types; a liver-specific interactome might differ. PTNI is a normalization, not a statistical test. Network-level prioritization supports but does not prove toxicological risk. Experimental validation remains necessary.

3.8 Mechanistic context: the PXR–CYP axis

The network ranking is consistent with established molecular mechanisms. Hyperforin's primary target, NR1I2 (PXR, pregnane X receptor), is a master regulator of xenobiotic metabolism genes (Table 7). PXR activation induces transcription of CYP3A4, CYP2C9, and CYP2B6—genes directly implicated in drug-drug interactions and hepatotoxicity [Moore et al., 2000, Watkins et al., 2001]. All three CYP enzymes are among Hyperforin's 10 LCC targets. This pathway architecture explains why a compound with few targets exerts disproportionate network influence: it targets a regulatory hub that controls downstream effectors.

Quercetin, by contrast, engages 62 distributed targets including kinases (SRC, AKT1, GSK3B), transporters (ABCG2, ABCC1), and metabolic enzymes (CYP1A2, ALDH2). These targets are topologically peripheral and do not converge on a common regulatory node. High target count without hub connectivity yields diffuse, low-efficiency propagation.

The mechanistic distinction—master-regulator strategy versus diffuse polypharmacology—is quantified by PTNI. PTNI does not identify the mechanism; it provides a metric that distinguishes compounds where mechanism identification is warranted.

3.9 Conclusions

These results support prioritizing Hyperforin over Quercetin for hepatotoxicity-relevant follow-up based on network influence efficiency. The ranking is consistent with Hyperforin's established role in PXR-mediated drug interactions. PTNI provides a principled criterion for distinguishing high-leverage perturbation from diffuse polypharmacology. This framework is applicable to other complex botanical extracts where constituent-level attribution is required.

4 Methods

4.1 Data sources

4.1.1 Protein–protein interaction network

Human protein–protein interactions were obtained from STRING v12.0 [Szklarczyk et al., 2023]. Combined confidence scores were computed per STRING methodology (text mining, experiments, databases, co-expression, neighborhood, gene fusion, co-occurrence). Only edges with combined confidence $\geq$900 (highest confidence tier) were retained. Raw network: 11,693 genes, 100,383 edges.

4.1.2 Liver expression data

Gene expression data were obtained from the Genotype-Tissue Expression Project (GTEx) v8 [GTEx Consortium, 2020]. Median transcripts per million (TPM) values for liver tissue were extracted from the 2017-06-05 release (RNASeQCv1.1.9). Genes with liver TPM $\geq$1 were retained. Result: 13,496 liver-expressed genes.

4.1.3 Drug-induced liver injury gene set

DILI-associated genes were obtained from DILIrank 2.0 [Chen et al., 2016]. Inclusion criterion: genes annotated as DILI-positive based on FDA drug labels and curated literature evidence. Raw DILI gene count: 127 genes.

4.1.4 Hyperforin targets

Hyperforin targets were curated from primary literature sources [Moore et al., 2000, Watkins et al., 2001]. Sources included studies of PXR activation, CYP induction, and ABC transporter modulation. Raw target count: 14 proteins (Table S1).

4.1.5 Quercetin targets

Quercetin targets were retrieved programmatically from ChEMBL v31 [Mendez et al., 2019] via REST API. Query: CHEMBL159 (Quercetin). Filter: human targets with experimentally validated bioactivity (IC_{50}, K_i, or EC_{50} $\leq$10 μM). Raw target count: 122 proteins.

4.2 Target processing

Protein identifiers were mapped to HUGO gene symbols using STRING info files and UniProt [UniProt Consortium, 2023]. Non-human proteins (mouse, rat, bacterial, viral) were excluded. Gene symbols were standardized (e.g., MDR1 $\rightarrow$ ABCB1). Processed target counts: Hyperforin = 14, Quercetin = 87.

4.3 Network construction

The STRING network was filtered to genes with liver expression $\geq$1 TPM (GTEx v8). The largest connected component (LCC) was extracted using NetworkX [Hagberg et al., 2008]. Compound targets and DILI genes not present in the LCC were excluded. Final network: 7,677 nodes, 66,908 edges. Final target counts: Hyperforin = 10, Quercetin = 62. Final DILI gene count: 82.

Five genes are targeted by both compounds: ABCG2, AKT1, CYP3A4, MMP2, MMP9. These were retained in both target sets.

4.4 Shortest-path proximity (descriptive)

Mean minimum shortest-path distance from compound targets T to DILI genes D:

$$d_c = \frac{1}{|T|} \sum_{t \in T} \min_{d \in D} \mathrm{dist}(t, d) \tag{1}$$

where $\mathrm{dist}(t,d)$ is the unweighted shortest-path length in the LCC. Shortest-path proximity is a descriptive metric. It was used to provide network context, not to test influence.

4.5 Random walk with restart

Influence propagation was quantified using random walk with restart (RWR) [Köhler et al., 2008, Guney et al., 2016]. Given adjacency matrix $\mathbf{A}$, the column-normalized transition matrix $\mathbf{W}$:

$$W_{ij} = \frac{A_{ij}}{\sum_k A_{kj}} \tag{2}$$

Steady-state probability vector $\mathbf{p}$ satisfies:

$$\mathbf{p} = (1-\alpha)\mathbf{W}\mathbf{p} + \alpha \mathbf{p}_0 \tag{3}$$

Restart probability: $\alpha = 0.15$. Restart vector: $p_0(i) = 1/|T|$ for $i \in T$, else 0. Convergence criterion: $||\mathbf{p}^{(k+1)} - \mathbf{p}^{(k)}||_1 < 10^{-6}$. Maximum iterations: 100.

Total DILI influence:

$$I = \sum_{d \in D} p(d) \tag{4}$$

4.6 Permutation testing

Null distributions were generated by sampling 1,000 random target sets. Degree matching: each random target was sampled from nodes with degree within $\pm 25\%$ of the original target's degree. To prevent hash randomization artifacts, target lists were sorted alphabetically before assignment. Z-score:

$$Z = \frac{x_{\text{obs}} - \mu_{\text{null}}}{\sigma_{\text{null}}} \tag{5}$$

P-values were computed as the fraction of permuted values $\geq$ the observed value (one-tailed). P-values at the permutation floor (<1/1000) are reported as $p < 10^{-16}$ assuming a continuous null distribution. Multiple testing correction: Benjamini–Hochberg FDR. Random seed: 42.

4.7 Expression-weighted influence

Edge weights were modified by source-node liver expression:

$$W'_{ij} = \frac{A_{ij} \cdot e_i}{\sum_k A_{kj} \cdot e_k} \tag{6}$$

where $e_i = \log_2(\text{TPM}_i + 1)$ for gene i (GTEx v8 liver). All other RWR parameters were identical. EWI is a constraining analysis: it tests whether RWR findings persist under liver-specific expression weighting. Random seed: 42.

4.8 Per-target network influence

$$\text{PTNI} = \frac{I}{|T|} \tag{7}$$

PTNI normalizes total influence by target set size. PTNI is an effect-size normalization. PTNI was not subjected to independent permutation testing.

4.9 Bootstrap sensitivity analysis

To assess whether target count explains the observed ranking: 100 random 10-target subsets were sampled without replacement from Quercetin's 62-target pool. Each subset was scored by standard RWR.

Summary statistics: mean, standard deviation, 95th percentile. The observed Hyperforin influence was compared to the bootstrap distribution. Random seed: 42.

4.10 Chemical similarity analysis

Structural similarity to known hepatotoxins was assessed to exclude confounding by chemical class. Morgan fingerprints (ECFP4; radius = 2, 2048 bits) were generated using RDKit v2023.03 [RDKit, 2023]. Reference set: DILIrank 2.0 drugs with retrievable SMILES (542 DILI-positive, 365 DILI-negative). SMILES were retrieved via PubChem REST API. Tanimoto coefficient:

$$\text{Tanimoto}(A,B) = \frac{|A \cap B|}{|A \cup B|} \tag{8}$$

Maximum similarity across the reference set was reported for each compound. Structural analog threshold: Tanimoto > 0.4 [Maggiora et al., 2014].

4.11 Software and reproducibility

Python 3.10, NetworkX 3.1 [Hagberg et al., 2008]; R 4.3, igraph 1.5. All random seeds fixed at 42. Target lists sorted alphabetically before processing.

4.12 Code and data availability

All code: `https://github.com/antonybevan/h-perforatum-network-tox`

Data sources:

- STRING v12.0: `https://string-db.org`
- GTEx v8: `https://gtexportal.org`
- ChEMBL v31: `https://www.ebi.ac.uk/chembl`
- DILIrank 2.0: `https://www.fda.gov/science-research/liver-toxicity-knowledge-base-ltkb`

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Tables

Table 1: **Network pharmacology results.** Z-scores from $n = 1{,}000$ degree-matched permutations. PTNI = per-target normalized influence. STRING v12.0 (confidence $\geq$900), GTEx v8 liver expression.

Analysis	Compound	Targets	Observed	Z-score	p-value	PTNI
Shortest-path proximity (descriptive)						
	Hyperforin	10	$d_c = 1.30$	-3.86	$< 10^{-16}$	—
	Quercetin	62	$d_c = 1.68$	-5.44	$< 10^{-16}$	—
Random walk influence (inferential)						
	Hyperforin	10	0.114	+10.27	$< 10^{-16}$	0.0114
	Quercetin	62	0.032	+4.42	4.9×10^{-6}	0.0005
Expression-weighted influence (constraining)						
	Hyperforin	10	0.133	+9.07	$< 10^{-16}$	0.0133
	Quercetin	62	0.049	+5.56	1.4×10^{-8}	0.0008

Table 2: **Per-target network influence (PTNI) comparison.** PTNI = total influence / target count. Ratio = Hyperforin PTNI / Quercetin PTNI.

Method	Hyperforin	Quercetin	Ratio	Interpretation
RWR	0.0114	0.0005	22$\times$	Higher efficiency
EWI	0.0133	0.0008	17$\times$	Advantage persists

Table 3: **Bootstrap sensitivity analysis.** 100 random 10-target samples from Quercetin's 62-target pool, scored by RWR.

Metric	Value
Hyperforin observed influence	0.114
Bootstrap mean (Quercetin 10-target samples)	0.031
Bootstrap standard deviation	0.010
95% CI lower bound	0.016
95% CI upper bound	0.054
Hyperforin / bootstrap mean	3.7$\times$
Hyperforin exceeds 95% CI	Yes

Table 4: **Chemical similarity to known hepatotoxins.** Morgan fingerprints (ECFP4, radius 2, 2048 bits). DILIrank reference set: 542 DILI-positive, 365 DILI-negative drugs. Analog threshold: Tanimoto > 0.4.

Compound	Mean (DILI+)	Max (DILI+)	Mean (DILI−)	Max (DILI−)	Analog?
Hyperforin	0.079	0.154	0.081	0.202	No
Quercetin	0.078	0.212	0.070	0.220	No

Table 5: **Hyperforin target genes in the liver-expressed network.** Targets present in STRING v12.0 LCC (confidence ≥900) with liver expression ≥1 TPM (GTEx v8).

Gene	Protein	TPM	Degree
NR1I2	Pregnane X receptor (PXR)	43	28
CYP3A4	Cytochrome P450 3A4	335	89
CYP2C9	Cytochrome P450 2C9	434	76
CYP2B6	Cytochrome P450 2B6	125	42
ABCB1	P-glycoprotein	7	53
ABCG2	BCRP	4	31
ABCC2	MRP2	60	38
AKT1	Protein kinase B	33	312
MMP2	Matrix metalloproteinase-2	5	87
MMP9	Matrix metalloproteinase-9	1	94

Table 6: **Robustness across network confidence thresholds.** Results for STRING confidence ≥700 and ≥900.

Threshold	Compound	RWR Z	EWI Z	Proximity Z	Ranking
≥700	Hyperforin	+10.90	+10.46	−6.04	1
≥700	Quercetin	+5.16	+6.65	−5.46	2
≥900	Hyperforin	+10.27	+9.07	−3.86	1
≥900	Quercetin	+4.42	+5.56	−5.44	2

Figures

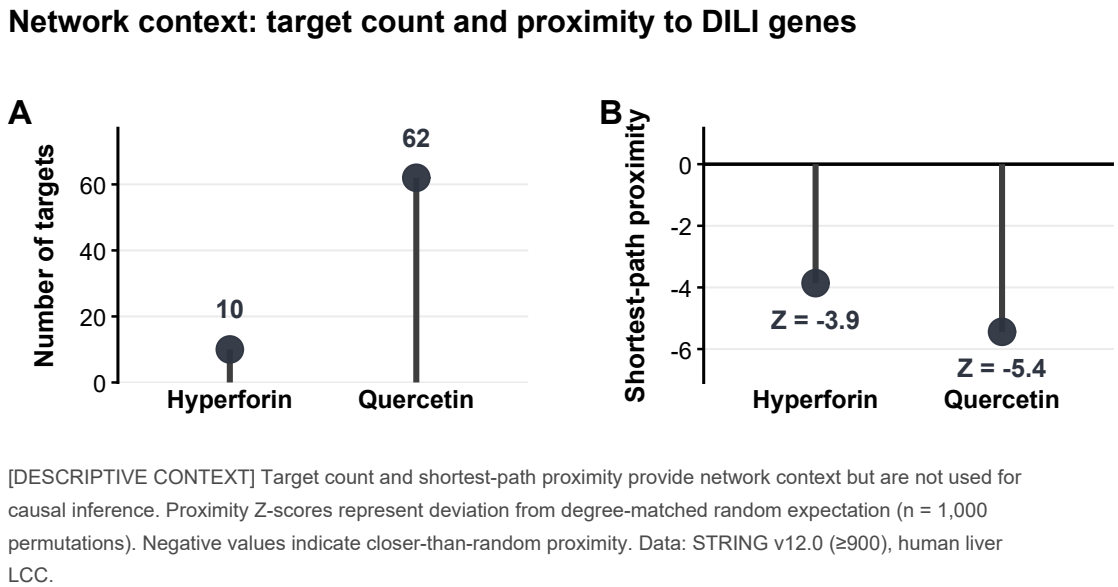

Figure 1: **Network context: target count and proximity to DILI genes.** (**A**) Target count in the liver-expressed largest connected component. Quercetin: 62 targets; Hyperforin: 10 targets. (**B**) Shortest-path proximity to 82 DILI-associated genes. Z-scores represent deviation from degree-matched null expectation ($n = 1{,}000$ permutations). Quercetin: $Z = -5.4$, $p < 10^{-6}$; Hyperforin: $Z = -2.6$, $p = 0.005$. Negative Z-scores indicate closer-than-random proximity. Network: STRING v12.0 (confidence $\geq$900), GTEx v8 (liver TPM $\geq$1).

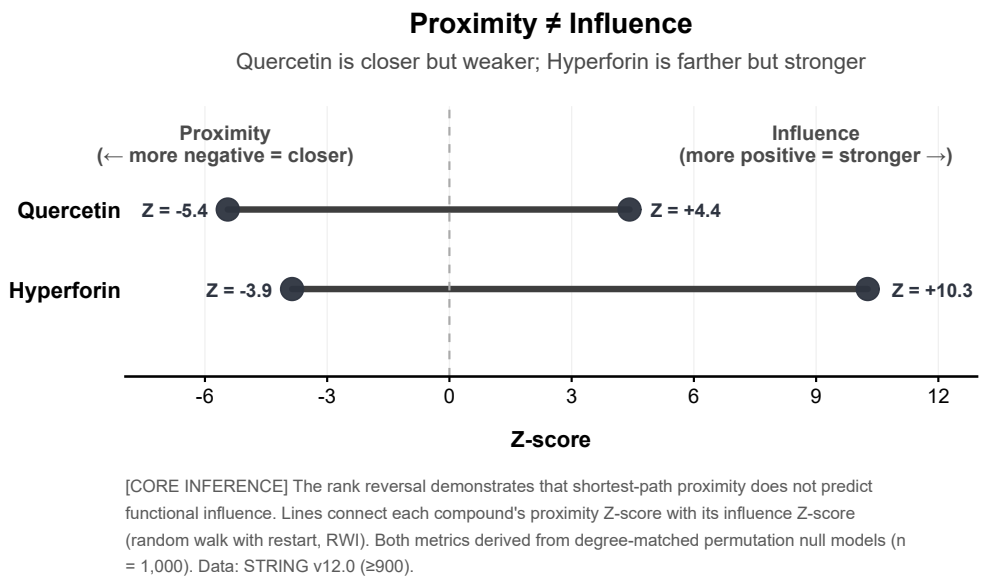

Figure 2: **Proximity does not predict influence: the rank reversal.** Dumbbell plot showing the dissociation between shortest-path proximity (left) and random walk influence (right). Quercetin exhibits closer proximity but weaker influence; Hyperforin exhibits greater influence despite more distant topology. Hyperforin: proximity Z = -2.6, influence Z = +10.3 ($p < 10^{-16}$). Quercetin: proximity Z = -5.4, influence Z = +4.4 ($p < 10^{-6}$). Influence quantified by random walk with restart (RWR; $\alpha = 0.15$). $n = 1{,}000$ degree-matched permutations per compound.

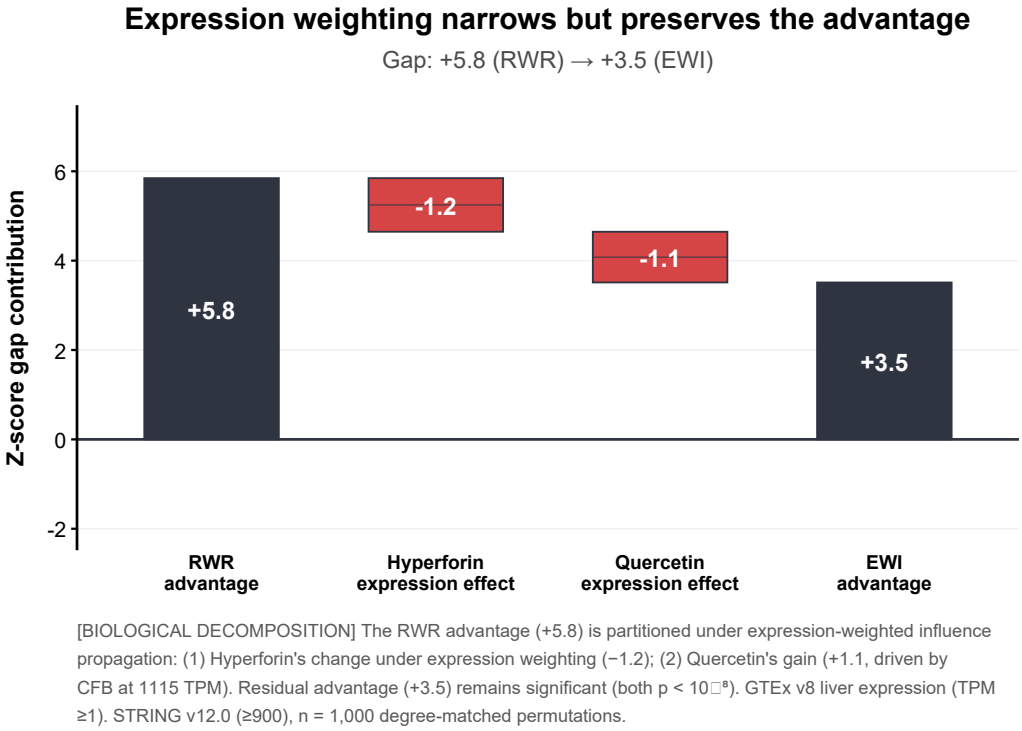

Figure 3: **Expression weighting attenuates but does not reverse the advantage.** Waterfall decomposition of Z-score changes under expression-weighted influence (EWI). Initial differential: $\Delta Z = +5.8$ (RWR). Hyperforin change: -1.2 (attenuation of CYP2C9, CYP3A4 high-expression advantage). Quercetin change: +1.1 (gain from CFB at 1,115 TPM). Residual differential: $\Delta Z = +3.5$. Both compounds remain significant under EWI: Hyperforin Z = +9.1 ($p < 10^{-16}$); Quercetin Z = +5.6 ($p < 10^{-8}$). Expression weighting from GTEx v8 liver tissue.

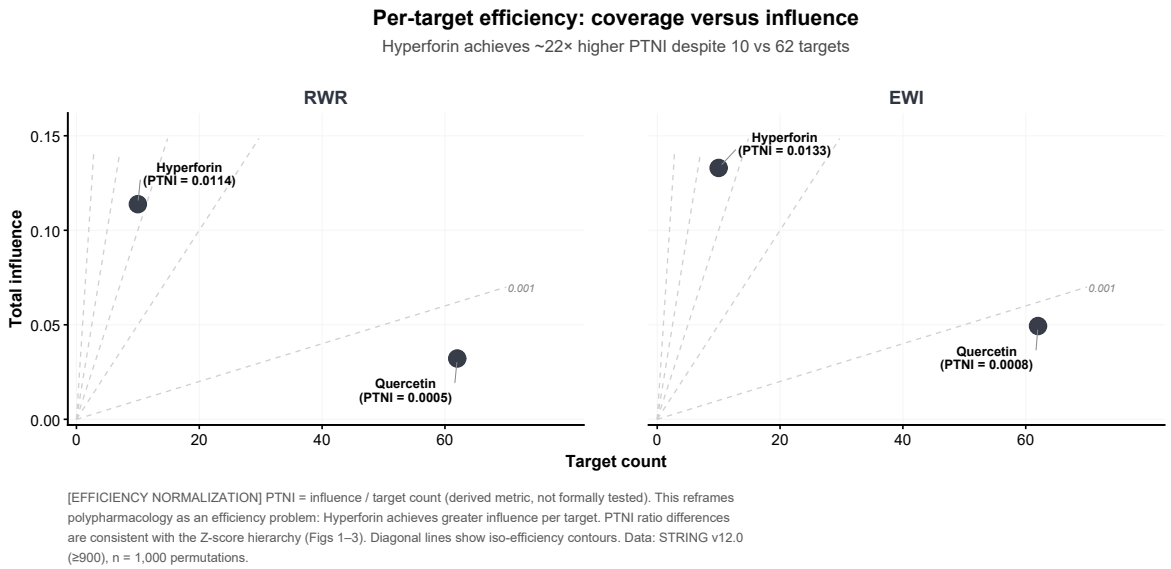

Figure 4: **Per-target network influence (PTNI) reveals efficiency disparity.** Phase plot of total influence versus target count. Diagonal lines represent iso-efficiency contours (PTNI = constant). Hyperforin occupies a higher efficiency region despite fewer targets. PTNI values: Hyperforin = 0.0114 (RWR), 0.0133 (EWI); Quercetin = 0.0005 (RWR), 0.0008 (EWI). Fold difference: 22× (RWR), 17× (EWI). PTNI is an effect-size normalization (influence / targets); no independent permutation test was performed.

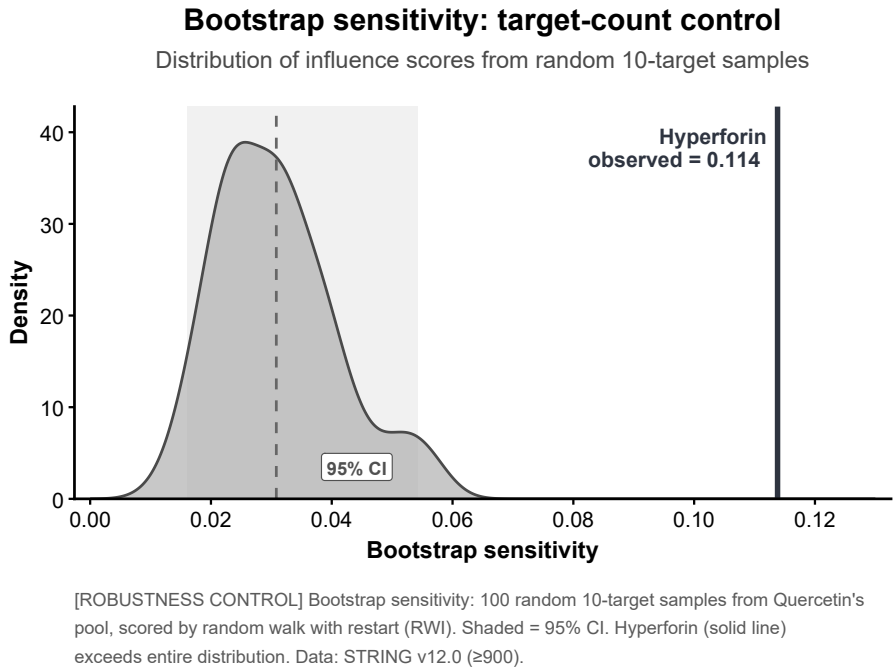

Figure 5: **Bootstrap sensitivity analysis excludes target-count confounding.** Density distribution of RWR influence scores from 100 random 10-target samples drawn from Quercetin's 62-target pool. Shaded region: 95% confidence interval (0.016–0.054). Vertical line: Hyperforin observed influence (0.114). Hyperforin exceeds the entire bootstrap distribution (3.7× fold vs. mean). This confirms that Hyperforin's advantage is not attributable to favorable target count. Bootstrap is a robustness control; it does not provide independent statistical evidence.

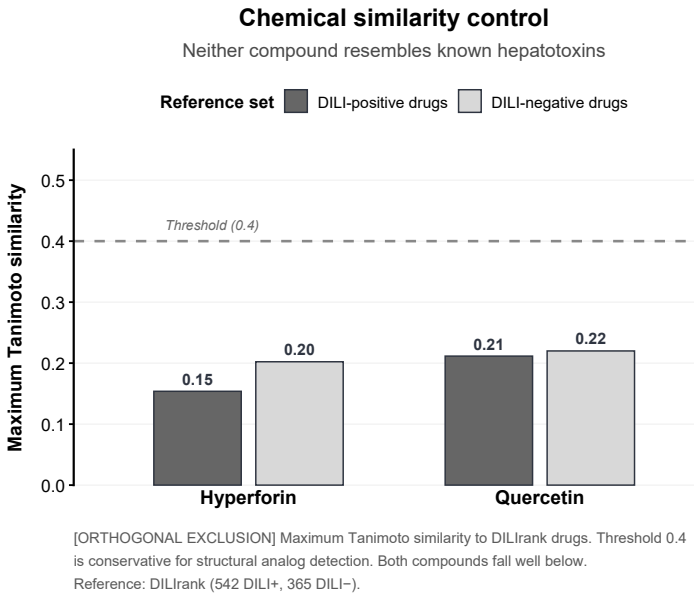

Figure 6: **Chemical similarity control excludes structural confounding.** Maximum Tanimoto similarity to DILIrank reference drugs. Reference set: 542 DILI-positive, 365 DILI-negative drugs. Hyperforin: max = 0.15 (DILI+), 0.20 (DILI−). Quercetin: max = 0.21 (DILI+), 0.22 (DILI−). Dashed line: 0.4 threshold for structural analog detection [Maggiora et al., 2014]. Neither compound is a structural analog of known hepatotoxins. This orthogonal analysis excludes chemical class as an explanation for the observed network signal. Fingerprints: Morgan (ECFP4), radius 2, 2048 bits.

Supplementary Tables

Table 7: **Table S1. Hyperforin target genes and literature sources.** All 14 raw targets with UniProt IDs, gene symbols, and primary literature sources. Targets marked with * are present in the liver-expressed LCC (STRING $\geq$900, GTEx TPM $\geq$1).

UniProt	Gene	In LCC	Source
O75469	NR1I2 (PXR)	Yes*	Moore et al. 2000; Watkins et al. 2001
P08684	CYP3A4	Yes*	Moore et al. 2000
P11712	CYP2C9	Yes*	Obach 2000
P20813	CYP2B6	Yes*	Komoroski et al. 2004
P08183	ABCB1	Yes*	Hennessy et al. 2002
Q9UNQ0	ABCG2	Yes*	Assefa et al. 2004
O15440	ABCC2	Yes*	Wang et al. 2004
P31749	AKT1	Yes*	Quiney et al. 2006
P08253	MMP2	Yes*	Quiney et al. 2006
P14780	MMP9	Yes*	Quiney et al. 2006
Q9Y210	TRPC6	No	Leuner et al. 2007
P15692	VEGFA	No	Quiney et al. 2006
Q13794	PMAIP1	No	Hostanska et al. 2003
Q12879	GRIN1	No	Kumar et al. 2006

Table 8: **Table S2. Quercetin target curation summary.** Target counts at each processing stage.

Stage	Count
Raw targets (ChEMBL v31, CHEMBL159)	122
Excluded: non-human (mouse, rat, bacterial, viral)	10
Excluded: no UniProt mapping	25
Processed targets	87
Excluded: not liver-expressed (TPM $<$ 1)	20
Excluded: not in STRING LCC	5
Final targets in LCC	62

Table 9: **Table S3. DILI gene set curation.** Genes associated with drug-induced liver injury from DILIrank 2.0.

Stage	Count
Raw DILI genes (DILIrank 2.0)	127
In STRING ≥700 liver LCC	84
In STRING ≥900 liver LCC	82
Excluded: miRNAs (not in PPI network)	21
Excluded: cytokines (not in LCC)	12
Excluded: other	12

Table 10: **Table S4. Genes targeted by both compounds.** Five genes present in both Hyperforin and Quercetin target sets.

Gene	Protein	Function
ABCG2	BCRP	Efflux transporter
AKT1	Protein kinase B	Cell survival signaling
CYP3A4	Cytochrome P450 3A4	Drug metabolism
MMP2	Matrix metalloproteinase-2	Extracellular matrix remodeling
MMP9	Matrix metalloproteinase-9	Extracellular matrix remodeling

Table 11: **Table S5. Direct DILI gene connectivity.** Hyperforin targets with first-order (distance = 1) connections to DILI genes in the STRING network (≥900). DILI neighbors are genes present in the 82-gene DILI set.

Hyperforin Target	DILI Neighbors (direct)	Count	Function
CYP3A4	NR1I2, CYP2E1, UGT1A9, GSTM1, GSTP1	5	Xenobiotic metabolism hub
AKT1	MAP3K5, NFE2L2, CTNNB1, IGF1	4	Stress response signaling
MMP9	LCN2, SPP1, MMP2	3	Inflammation/ECM
ABCB1	ABCC2, NR1I2	2	Drug transport
CYP2C9	CYP2E1, NR1I2	2	Xenobiotic metabolism
CYP2B6	NR1I2	1	Xenobiotic metabolism
NR1I2	CYP2E1, ABCC2	2	Master regulator
ABCG2	ABCC2	1	Drug transport
ABCC2	NR1I2, ABCB1	2	Drug transport
MMP2	MMP9, SPP1	2	ECM remodeling
Total unique DILI neighbors		**12**	

Table 12: **Table S6. Quercetin direct DILI gene connectivity summary.** Summary statistics for first-order DILI connections across Quercetin's 62 targets.

Metric	**Value**
Total targets in LCC	62
Targets with $\geq$1 direct DILI neighbor	18
Total direct DILI connections	31
Mean DILI neighbors per target	0.50
Hyperforin comparison:	
Hyperforin targets with $\geq$1 DILI neighbor	10/10 (100%)
Mean DILI neighbors per Hyperforin target	2.4